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Notes on the total positivity of the compression of double Riordan arrays

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Let $\widehat{R} = (\hat{d}(t); \hat{h}_1(t), \hat{h}_2(t))$ be the compression of a double Riordan array. We show that if $\hat{d}(t), \hat{h}_1(t)$ and $\hat{h}_2(t)$ are all Pólya frequency formal power series, then $\widehat{R}$ is totally positive.

Keywords: Double Riordan arrays; total positivity; Pólya frequency sequence.

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1. Introduction

Riordan arrays play an important unifying role in enumerative combinatorics [12, 15]. Let $d(t) = \sum_{n\geq 0} d_n t^n$ and $h(t) = \sum_{n\geq 0} h_n t^n$ be two formal power series. A *Riordan array*, denoted by $(d(t), h(t))$, is an infinite matrix whose generating function of the kth column is $d(t)h^k(t)$ for $k \geq 0$. We say that a Riordan array $(d(t), h(t))$ is *proper* if $d_0 = 1, h_0 = 0$ and $h_1 \neq 0$. It is well known [15] that the set of proper Riordan arrays is a group under the matrix multiplication and

$$(d(t), h(t)) \cdot (g(t), f(t)) = (d(t)g(h(t)), f(h(t))).$$

The identity matrix can be written as $(1, t)$ and the inverse of $(g(t), f(t))$ is given by

$$(1/g(\bar{f}(t)), \bar{f}(t)),$$

where $\bar{f}$ is the compositional inverse of f, i.e. $f(\bar{f}(t)) = \bar{f}(f(t)) = t$.

The double Riordan array, introduced by Davenport *et al.* in [7], has been extensively studied for its sequence characterization [1, 8], q-analogue representation, and matrix product formulas [19], as well as for deriving expressions and matrix properties of its halves using the compression method [18]. Let $d(t) = \sum_{k\geq 0} d_{2k} t^{2k}$, $h_1(t) = \sum_{k\geq 0} h_{1,2k+1} t^{2k+1}$ and $h_2(t) = \sum_{k\geq 0} h_{2,2k+1} t^{2k+1}$, where $d(t)$ is an even formal power series, h_1 and h_2 are odd formal power series, and $d_0, h_{1,1}, h_{2,1} \neq 0$. Then a *double Riordan array* of d, h_1 and h_2, denoted by $(d; h_1, h_2)$, has column vectors $(d, dh_1, dh_1h_2, dh_1^2h_2, dh_1^2h_2^2, \ldots)$. It is well known [7] that the set of double Riordan arrays is a group under the matrix multiplication and

$$(d; h_1, h_2)(g; f_1, f_2) = (dg(\sqrt{h_1h_2}); \sqrt{h_1/h_2} f_1(\sqrt{h_1h_2}), \sqrt{h_2/h_1} f_2(\sqrt{h_1h_2})).$$

The identity matrix can be written as $(1; t, t)$ and let $f = \sqrt{h_1h_2}$, the inverse of $(d; h_1, h_2)$ is given by

$$(1/d(\bar{f}); t\bar{f}/h_1(\bar{f}), t\bar{f}/h_2(\bar{f})),$$

where $\bar{f}$ is the compositional inverse of f, i.e. $f(\bar{f}(t)) = \bar{f}(f(t)) = t$.

Let $[d_{n,k}]_{n\geq k\geq 0} = (d; h_1, h_2)$ be a double Riordan array, and its *compression* $[\hat{d}_{n,k}]_{n,k\geq 0}$ [8] is defined by

$$\hat{d}_{n,k} := d_{2n-k,k}, \quad n \geq k \geq 0, \tag{1.1}$$

that is

$$\hat{d}_{n,k} = \begin{cases} [t^n]\hat{d}(\hat{h}_1\hat{h}_2)^{k/2}, & \text{if } k \text{ is even,} \\ [t^n]\hat{d}\hat{h}_1(\hat{h}_1\hat{h}_2)^{(k-1)/2}, & \text{if } k \text{ is odd,} \end{cases}$$

where

$$\hat{d}(t) = \sum_{k\geq 0} d_{2k} t^k, \quad \hat{h}_1(t) = \sum_{k\geq 0} h_{1,2k+1} t^{k+1} \quad \text{and} \quad \hat{h}_2(t) = \sum_{k\geq 0} h_{2,2k+1} t^{k+1}.$$

As an example, let us consider the Fibonacci tree F shown in Stanley [17], which is defined to be the rooted tree with root v, such that the root has degree one, the

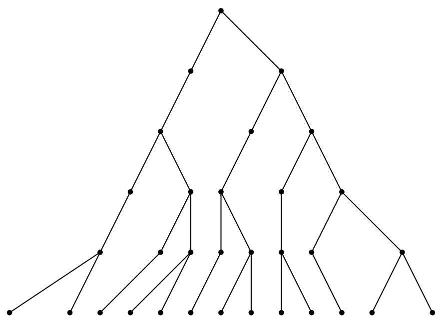

Fig. 1. The Fibonacci tree.

child of every vertex of degree one has degree two, and the two children of every vertex of degree two have degrees one and two.

The Pascal–Fibonacci array [8]:

$$\widehat{P} = \left(\frac{1}{1-t}; t, \frac{t}{1-t}\right) = \begin{bmatrix} 1 & & & & & & \\ 1 & 1 & & & & & \\ 1 & 1 & 1 & & & & \\ 1 & 1 & 2 & 1 & & & \\ 1 & 1 & 3 & 2 & 1 & & \\ 1 & 1 & 4 & 3 & 3 & 1 & \\ \vdots & & & & & & \ddots \end{bmatrix}$$

is the compression of the double Riordan array $(1/(1-t^2); t, 1/(1-t^2))$, which represents the Fibonacci tree shown in Fig. 1. The row sums of $\widehat{P}$ are Fibonacci numbers.

Multiplication rule holds among Riordan arrays and the compression of the double Riordan arrays. For instance, we have the following lemma.

Lemma 1.1 ([8, Proposition 3.10]). *Let (g, f) and $(d; h_1, h_2)$ be a Riordan array and a double Riordan array, respectively. Then the compression of $(d; h_1, h_2)$, denoted by $(\hat{d}; \hat{h}_1, \hat{h}_2)$, satisfies $\hat{d}(t) = d(\sqrt{t}), \hat{h}_1(t) = \sqrt{t}h_1(\sqrt{t})$, and $\hat{h}_2(t) = \sqrt{t}h_2(\sqrt{t})$. The multiplication of (g, f) and $(\hat{d}; \hat{h}_1, \hat{h}_2)$ is also a compression of a double Riordan array. More precisely,*

$$(g, f)(\hat{d}; \hat{h}_1, \hat{h}_2) = (g\hat{d}(f); \hat{h}_1(f), \hat{h}_2(f)). \tag{1.2}$$

Total positivity theory has been widely studied in both analysis and combinatorics [2–4, 10, 11, 14, 20]. Following Karlin [10], a (finite or infinite) matrix is called *totally positive* (or shortly, TP) if its minors of all orders are non-negative. Let $(a_n)_{n\geq 0}$ be an infinite sequence of non-negative numbers. Define its Toeplitz matrix

$$[a_{i-j}]_{i,j\geq 0} = \begin{bmatrix} a_0 & & & & \\ a_1 & a_0 & & & \\ a_2 & a_1 & a_0 & & \\ a_3 & a_2 & a_1 & a_0 & \\ \vdots & & & & \ddots \end{bmatrix}.$$

We say that the sequence is a *Pólya frequency sequence* (PF) if the corresponding Toeplitz matrix is TP. We identify a finite sequence $a_0, a_1, \ldots, a_n$ with the infinite sequence $a_0, a_1, \ldots, a_n, 0, 0, \ldots$. A fundamental characterization for PF sequences is due to Schoenberg and Edrei, which states that a sequence $(a_n)_{n\geq 0}$ of non-negative numbers is PF if and only if its generating function has the form

$$\sum_{n\geq 0} a_n t^n = \lambda t^m e^{\gamma t} \frac{\prod_{j\geq 0}(1+\alpha_j t)}{\prod_{j\geq 0}(1-\beta_j t)}, \tag{1.3}$$

where $\lambda > 0, m \in \mathbb{N}, \alpha_j, \beta_j, \gamma \geq 0$ and $\sum_{j\geq 0}(\alpha_j + \beta_j) < +\infty$ (see [10, p. 412] for instance). In this case, the generating function is called a *PF formal power series* (see [2, 14] for instance). In recent years, many researchers focused on the total positivity of the Riordan arrays (see [5, 6, 13] for instance). Our concern in this paper is the total positivity of the compression of the double Riordan arrays.

In the next section we present our main result, and then apply it to the Pascal–Fibonacci array and a compression of a double Riordan array defined by He. In Sec. 3, we propose some problems for further work.

2. Main Results and Applications

Theorem 2.1. *Let $R = (d; h_1, h_2)$ be a double Riordan array, and let its compression is $\widehat{R} = (\hat{d}; \hat{h}_1, \hat{h}_2)$. If $\hat{d}, \hat{h}_1$ and $\hat{h}_2$ all are PF formal power series, then $\widehat{R}$ is TP.*

Proof. By Lemma 1.1, each the compression of the double Riordan arrays have the decomposition:

$$(\hat{d}; \hat{h}_1, \hat{h}_2) = (\hat{d}, t)(1; \hat{h}_1, \hat{h}_2). \tag{2.1}$$

By the classic Cauchy–Binet formula, the product of two TP matrices is also TP. So, the total positivity of both $(\hat{d}, t)$ and $(1; \hat{h}_1, \hat{h}_2)$ implies that of $(\hat{d}; \hat{h}_1, \hat{h}_2)$. We need to prove the following two statements.

(i) If the sequence $(\hat{d}_n)_{n\geq 0}$ is PF, then the corresponding Toeplitz array $(\hat{d}, t)$ is TP.
(ii) If the sequence $(\hat{h}_{1,n})_{n\geq 1}$ and $(\hat{h}_{2,n})_{n\geq 1}$ are PF, then the corresponding array $(1; \hat{h}_1, \hat{h}_2)$ is TP.

The first statement is clear by the definition, since $(\hat{d}, t)$ is precisely the Toeplitz matrix of the sequence $(\hat{d}_n)_{n\geq 0}$. It remains to prove the second statement. Note that

$$(1, \hat{h}_1, \hat{h}_2) = \begin{bmatrix} 1 & \\ & (\hat{h}_1/t; \hat{h}_2, \hat{h}_1) \end{bmatrix}.$$

Hence the total positivity of $(1; \hat{h}_1,\ \hat{h}_2)$ is equivalent to that of $(\hat{h}_1/t; \hat{h}_2, \hat{h}_1)$. So it suffices to prove the following statement.

(iii) If the sequence $(\hat{h}_{1,n})_{n\geq 1}$ and $(\hat{h}_{2,n})_{n\geq 1}$ are PF, then the corresponding array $(\hat{h}_1/t; \hat{h}_2, \hat{h}_1)$ is TP. Likewise,

$$(\hat{h}_1/t; \hat{h}_2, \hat{h}_1) = (\hat{h}_1/t, t)(1; \hat{h}_2, \hat{h}_1). \tag{2.2}$$

So, the total positivity of both $(\hat{h}_1/t, t)$ and $(1; \hat{h}_2, \hat{h}_1)$ implies that of $(\hat{h}_1/t; \hat{h}_2, \hat{h}_1)$. We need to go ahead and prove the following two statements.

(iv) If the sequence $(\hat{h}_{1,n})_{n\geq 1}$ is PF, then the corresponding Toeplitz array $(\hat{h}_1/t, t)$ is TP.

(v) If the sequence $(\hat{h}_{1,n})_{n\geq 1}$ and $(\hat{h}_{2,n})_{n\geq 1}$ are PF, then the corresponding array $(1; \hat{h}_2, \hat{h}_1)$ is TP.

The fourth statement is clear by the definition, since $(\hat{h}_1/t, t)$ is precisely the Toeplitz matrix of the sequence $(\hat{h}_{1,n})_{n\geq 1}$. It remains to prove the second statement. We continue to use the same method

$$(1; \hat{h}_2, \hat{h}_1) = \begin{bmatrix} 1 & \\ & (\hat{h}_2/t; \hat{h}_1, \hat{h}_2) \end{bmatrix}.$$

Hence the total positivity of $(1; \hat{h}_2, \hat{h}_1)$ is equivalent to that of $(\hat{h}_2/t; \hat{h}_1, \hat{h}_2)$. So it suffices to prove the following statement.

(vi) If the sequence $(\hat{h}_{1,n})_{n\geq 1}$ and $(\hat{h}_{2,n})_{n\geq 1}$ are PF, then the corresponding array $(\hat{h}_2/t; \hat{h}_1, \hat{h}_2)$ is TP. By(2.1), we have

$$(\hat{h}_2/t; \hat{h}_1, \hat{h}_2) = (\hat{h}_2/t, t)(1; \hat{h}_1, \hat{h}_2). \tag{2.3}$$

Let

$$H_1 = (\hat{h}_1/t; \hat{h}_2, \hat{h}_1), \quad H_2 = (\hat{h}_2/t; \hat{h}_1, \hat{h}_2), \quad T_1 = (\hat{h}_1/t, t), \quad T_2 = (\hat{h}_2/t, t),$$

then by (2.2) and (2.3), we have

$$H_1 = T_1 \begin{bmatrix} 1 & \\ & H_2 \end{bmatrix}, \quad H_2 = T_2 \begin{bmatrix} 1 & \\ & H_1 \end{bmatrix}. \tag{2.4}$$

Let $H_1[n]$ be the submatrix consisting of the first n columns of H_1. Clearly, H is TP if and only if its submatrices $H_1[n]$ are all TP. Thus it suffices to show that

$H_1[n]$ is TP for all $n \geq 1$. We proceed by induction on n. By (2.4), we have

$$H_1[n+2] = T_1 \begin{bmatrix} 1 & \\ & H_2[n+1] \end{bmatrix}, \quad H_2[n+1] = T_2 \begin{bmatrix} 1 & \\ & H_1[n] \end{bmatrix}. \tag{2.5}$$

Clearly, T_1 and T_2 are TP since they are the Toeplitz matrix of the PF sequence $(\hat{h}_{1,n})_{n\geq 1}$ and $(\hat{h}_{2,n})_{n\geq 1}$, respectively. Now, if $H_1[n]$ is TP, then so is $\begin{bmatrix} 1 & \\ & H_1[n] \end{bmatrix}$. Thus the product $H_2[n+1]$ is also TP by (2.5), as desired. Further, the product $H_1[n+2]$ is also TP by (2.5). □

Remark 2.2. If $\hat{h}_1 = \hat{h}_2 = \hat{h}$, then $(\hat{d}; \hat{h}_1, \hat{h}_2) = (\hat{d}, \hat{h})$ is also a Riordan array, and furthermore, if both $\hat{d}(t)$ and $\hat{h}(t)$ are Pólya frequency formal power series, then the Riordan array $(\hat{d}, \hat{h})$ is TP. This result is also discussed in [6].

Example 2.3. Consider the compression of a double Riordan array $(\frac{1}{1-t}; \frac{t}{1-t}, t)$, or

$$\left(\frac{1}{1-t}; \frac{t}{1-t}, t\right) = \begin{bmatrix} 1 & & & & & & \\ 1 & 1 & & & & & \\ 1 & 2 & 1 & & & & \\ 1 & 3 & 2 & 1 & & & \\ 1 & 4 & 3 & 3 & 1 & & \\ 1 & 5 & 4 & 6 & 3 & 1 & \\ \vdots & & & & & & \ddots \end{bmatrix}$$

is TP by Theorem 2.1. We see that the alternating row sums of the matrix are $F_n - 1$, where F_n is the nth Fibonacci numbers, see also Sloane's OEIS [16, A194005].

Example 2.4. The compression of a double Riordan array $(\frac{1}{1-2t}; t, \frac{t}{1-2t})$, or

$$\left(\frac{1}{1-2t}; t, \frac{t}{1-2t}\right) = \begin{bmatrix} 1 & & & & & & \\ 2 & 1 & & & & & \\ 4 & 2 & 1 & & & & \\ 8 & 4 & 4 & 1 & & & \\ 16 & 8 & 12 & 4 & 1 & & \\ 32 & 16 & 32 & 12 & 6 & 1 & \\ \vdots & & & & & & \ddots \end{bmatrix}$$

is TP by Theorem 2.1. We can see that this matrix associated to the Chebyshev polynomials of the fourth kind (see [16, A228565 and A180870]).

It follows immediately from Theorem 2.1 that the Pascal–Fibonacci array $\widehat{P} = (\frac{1}{1-t}; t, \frac{t}{1-t})$ is TP. More generally, consider the compression of the double Riordan array defined by He [8]

$$\widehat{M} = \left(\frac{t^m}{(1-t)^{m+1}}; \frac{t^{b-a+1}}{(1-t)^b}, \frac{t^{c-d+1}}{(1-t)^d}\right),$$

where a, b, c, d and $m \in \mathbb{N}$.

Corollary 2.5. *If $b - a \geq 0$ and $c - d \geq 0$, then $\widehat{M}$ is TP.*

3. Remarks

It is well known [9] that a proper Riordan array $R = [r_{n,k}]_{n,k\geq 0} = (g(t), f(t))$ can be characterized by two sequences $(a_n)_{n\geq 0}$ and $(z_n)_{n\geq 0}$ such that

$$r_{0,0} = 1, \quad r_{n+1,0} = \sum_{j\geq 0} z_j r_{n,j}, \quad r_{n+1,k+1} = \sum_{j\geq 0} a_j r_{n,k+j}$$

for $n, k \geq 0$. X. Chen *et al.* [5, Theorem 2.1] gave a criterion for the total positivity of Riordan arrays by means of A- and Z-sequences: if the matrix

$$J(R) = \begin{bmatrix} z_0 & a_0 & & & \\ z_1 & a_1 & a_0 & & \\ z_2 & a_2 & a_1 & a_0 & \\ \vdots & \vdots & & & \ddots \end{bmatrix}$$

is TP, then so is the Riordan array R. It is natural to ask for similar conclusions of the compression of the double Riordan array. However, we do not know how to derive the $\widehat{A}_1$-, $\widehat{A}_2$- and $\widehat{Z}$-sequences for the compressed of a double Riordan array, or whether these sequences can be accurately expressed.

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